

DIRECT FARM-TO-CONSUMER MARKETPLACE & ANTI-MIDDLEMAN GATEKEEPER

Client Frontend React 19 + Tailwind v4 Vite 8 Build Engine	Database Engine Neon Serverless PG Auto-Pooled & SSL Secured	Real-Time Telemetry Server-Sent Events Live GPS Cold-Chain	Anti-Middleman Gate PM-Kisan & Bhulekh Govt Land Verified KYC
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1. Frontend & Client-Side Presentation Layer

Technology	Version / Package	Functional Role & Architectural Purpose
React 19	v19.2.8	Component architecture utilizing modern concurrent rendering, custom hooks, and unified atomic context for Farmer, Buyer, and Admin gateways.
Vite 8	v8.2.2	Ultra-fast frontend dev server and production builder with integrated custom API middleware plugins.
Tailwind CSS v4	v4.3.3 (@tailwindcss/vite)	Next-gen CSS utility engine delivering custom glassmorphic cards, responsive mobile-first layouts, and high-contrast color tokens.
Recharts	v3.10.1	Visualizes dynamic AI demand forecasting curves, seasonal mandi vs. direct price spreads, and farmer revenue gains.
Leaflet	v1.9.4	Renders interactive geospatial maps, cold-chain GPS telemetry markers, and optimized multi-stop transit corridors.
Lucide React	v1.37.0	Accessible, modern SVG iconography across product cards, order timelines, and dashboard controls.

2. Backend & Serverless Middleware Layer

Component	Implementation	Architectural Details
Vite API Middleware	Node.js ES Modules	Lightweight HTTP API gateway handling authentication endpoints, product catalog queries, and zero-middlemen escrow payouts.
Server-Sent Events (SSE)	Text/Event-Stream	One-way real-time push streaming cold-chain vehicle GPS updates, temperature alerts, and order milestones to buyers without polling.
Spring Boot Microservice	Java 17 / Spring Boot 3	Alternative enterprise backend (in backend/) with Spring Security, JPA/Hibernate, and JWT authentication.

3. Cloud Database & Storage Layer (Neon PostgreSQL)

Schema Table	Key Columns	Purpose & Security Enforcement
govt_verified_farmers	farmer_id, khasra_no, aadhaar, kcc, land_acres, phone	Government-certified land registry. Strictly rejects unregistered middlemen attempting to register as farmers.
admins	admin_id, full_name, phone, designation, zone, access_tier	Operational command directory. Dual-key (Admin ID + Registered Phone) gatekeeper for fleet operations.
otp_records	phone, email, otp, role, verified, created_at	Manages dynamic 6-digit OTP lifecycles with expiration and verification status tracking.
products & orders	id, price_per_kg, available_qty, origin_json, telemetry	Real-time crop listings and active multi-step orders with GPS coordinates and driver telemetry.
payouts	order_id, utr, farmer_upi, amount, zero_middlemen_cut	Tracks instant UPI settlement transfers with verified UTR bank transaction IDs.

4. Anti-Middleman Security & Regulatory Architecture

- **Anti-Middleman Bhulekh KYC:** Farmers cannot list produce without validating their Farmer ID, Aadhaar, and Khasra title against official state records.
- **Zero Public Admin Registration:** The logistics gateway is closed to public signups; only pre-approved officers in admins can log in via Phone + Admin ID + OTP.
- **Trustless Escrow Settlement:** Buyer payments are locked in an escrow state until digital delivery confirmation triggers instant UPI payout.